



Vortex-induced vibration effects on train dynamics and safety-comfort evaluation in wind-train-bridge coupled system

Tianyu Chen^{a,b}, Cunming Ma^{a,b,*}, Qingsong Duan^c, Kai Cheng^{a,b}, Cheng Pei^{d,e}

^a Department of Bridge Engineering, School of Civil Engineering, Southwest Jiaotong University, Chengdu, 610031, China

^b Key Laboratory for Wind Engineering of Sichuan Province, Chengdu, 610031, China

^c School of Civil Engineering and Architecture, Southwest University of Science and Technology, Mianyang, 621010, China

^d School of Airport Engineering, Civil Aviation Flight University of China, Guanghan, 618307, China

^e Sichuan Engineering Research Center for Smart Operation and Maintenance of Civil Aviation Airports, Guanghan, 618307, China

ARTICLE INFO

Keywords:

Vortex-induced vibration
High-speed railway
Wind tunnel test
Wind-train-bridge coupling vibration
Dynamic response

ABSTRACT

With the continuous improvement of the bridge span, the phenomenon of vortex-induced vibration (VIV) in railway bridges can disrupt the normal functioning of railway systems. This paper establishes a method for wind-train-bridge-VIV, focusing on the dynamic responses of the bridge and the impact of the train during such events. Additionally, the study evaluates the effects of critical parameters like the amplitude of VIV, the train's operational velocity, and the initial phase angle of VIV on the dynamic reactions of the train and the bridge. The allowable values of VIV for the railway bridge are also investigated. The findings reveal that the three key factors substantially impact the dynamic reactions experienced by the railway bridge and the train. Notably, the VIV amplitude is a dominant factor in the vertical dynamic response, exerting a substantial impact on train ride comfort. Consequently, it is imperative to impose limitations on the VIV amplitude in railway bridges to maintain operational efficiency and passenger comfort.

1. Introduction

The proliferation of high-speed rail networks has highlighted the issue of aerodynamic challenges faced by long-span railway bridges, which are crucial for maintaining uninterrupted train service. The crosswind may lead to vortex-induced vibration (VIV) of the railway bridge, potentially leading to track alignment modifications and, consequently, affecting passenger comfort. The reciprocal effect of the train's motion on the bridge's vibration further complicates the system. Collectively, these elements form a wind-train-bridge-VIV system, which must be thoroughly understood and addressed to ensure the safe and efficient operation of high-speed railways.

The coupling vibration system involving wind, train, and bridge is typically dissected by establishing the interconnections among these three elements. Ji (Ji et al., 2018) proposed a method that integrates the flexible track function embedded within SIMPACK software with flexible track slab data imported from ANSYS, effectively realizing the coupling of the train-flexible track. Han (Han et al., 2020) employed a synergistic application of finite element analysis and SIMPACK software to construct a wind-train-track-bridge coupling model, thereby

advancing the understanding of the complex interactions within this system. Subject to the influence of fluctuating aerodynamic forces, the safety and comfort of the trains are likely to decrease. Considering the influence of unsteady aerodynamic forces, both safety and comfort can be evaluated in a more conservative way. Wang (Wang et al., 2021) developed a model for wind-train-bridge interactions, incorporating the train's longitudinal degree of freedom, and examined how wind and train speeds affect the vehicle's response. The results indicate that the wind generated by the train significantly impacts the vehicle's dynamic reactions, while its influence on the bridge's response is minimal. These studies have significantly advanced our understanding of the wind-train-bridge coupled system. However, as the span of railway bridges increases and their frequency decreases, long-span railway bridges are more susceptible to VIVs. Existing models fail to comprehensively incorporate the impacts of VIV, a critical aspect that requires further investigation.

As the main span of railway bridges increased, it became imperative to examine the VIV. The most straightforward approach to investigating these VIV characteristics is through wind tunnel testing. Many scholars have estimated the VIV characteristics through computational fluid

* Corresponding author. Department of Bridge Engineering, School of Civil Engineering, Southwest Jiaotong University, Chengdu, 610031, China
E-mail address: mcm@swjtu.edu.cn (C. Ma).

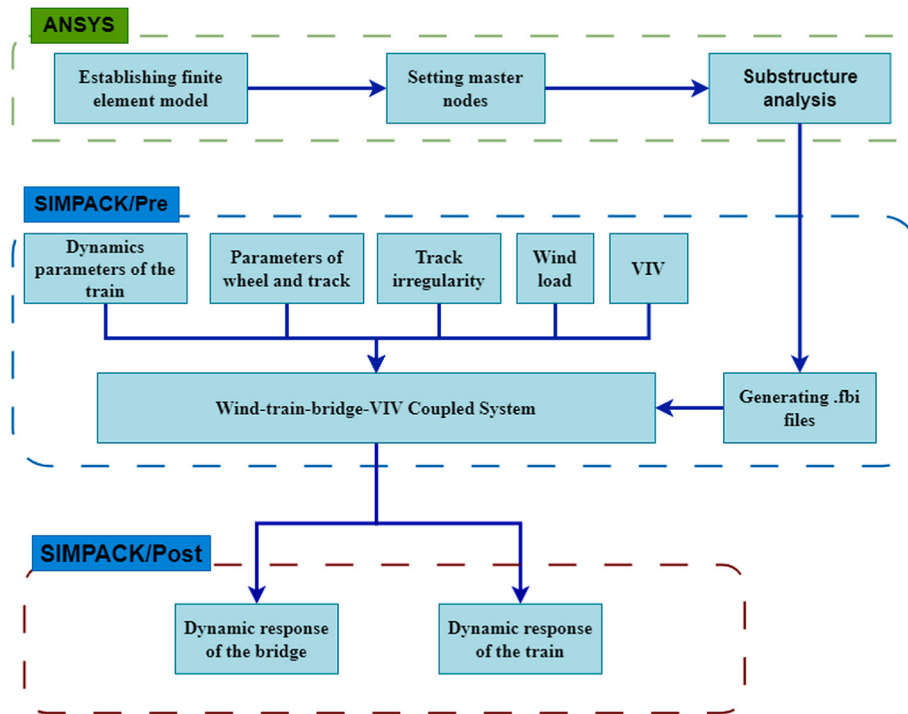


Fig. 1. Flow chart of wind-train-bridge-VIV coupled system.

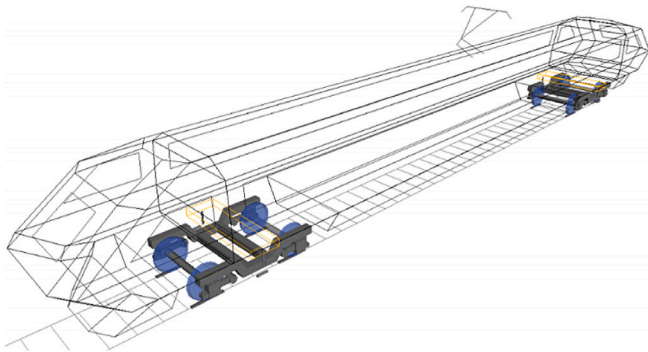


Fig. 2. Structure of the CRH2 train.

dynamics (CFD) simulations and wind tunnel tests (Noguchi et al., 2020, Wu et al., 2023; Yan et al., 2022, Li et al., 2024). Duan (Duan et al., 2023) explored the VIV characteristics of a narrow-streamlined box girder railway bridge and analyzed how the wind attack angle and the damping ratio affect the girder's VIV response. The results indicate that the base of the railing is responsible for inducing vertical VIV. Chen (Chen et al., 2022) conducted a comprehensive analysis of how gap arrangements affect the VIV of the twin-box girder using wind tunnel experiments, and also explored the aerodynamic features of such structures through CFD simulations. Vortex formation at the gap of the twin-box girder is the primary cause of VIV. The shedding of these vortices generates periodic vortex-induced forces, causing the twin-box girder to vibrate. In return, the vibration affects vortex formation and development, creating a fluid-structure interaction.

The allowable values for VIV directly inform the wind-resistant design of railway bridges and are of significant importance. This necessitates further research into the coupled vibrations of bridges, trains, and wind under conditions of VIV, as this area has seen relatively scant investigation to date. At present, there are more established regulations regarding the allowable values for VIV on highway bridges. Some scholars have formulated the VIV amplitude limit from the aspects of

structural fatigue, human comfort, bridge-bearing capacity, and drivers' vision (Xu, Guo et al., 2004; Zhou and Chen, 2016; Yu et al., 2020; Fu et al., 2025). Zhu (Zhu et al., 2021) proposed a method to assess the comfort of bridges under random traffic flow through VIV. This method fully considers traffic patterns and the interaction between wind, bridge, and traffic flow. Results show that highway surface roughness is the main factor affecting passenger comfort. Despite extensive research on the VIV of highway bridges, railway bridges have received comparatively less attention in this regard.

This study concentrates on investigating the VIV characteristics of the box girder in a cable-stayed bridge, which features a main span of 335m. An integrated wind-train-bridge coupled model, incorporating the effects of VIV, is meticulously constructed. This model accounts for the VIV influence on the bridge by introducing it through displacements, allowing for a nuanced exploration of the system's dynamic behavior under the impact of such vibrations. Building on this analysis, the study further determines the threshold for VIV amplitude for the bridge, ensuring operational safety and ride comfort for the train. This research contributes to the understanding of VIV in railway bridges and provides guidelines for their design and operational considerations.

2. Wind-train-bridge-VIV coupling vibration system

The dummy coupling technique is employed to achieve the coupling between the train and the bridge. The dummy element serves as an intermediary, connecting the flexible bridge to the train, thus creating a composite rigid-flexible coupling model. Since the mass of the dummy element can be ignored for the whole system, its impact on the wheel-rail forces transmitted downward can also be ignored (Li et al., 2018). In the simulation model, the interaction between the rigid wheel and the dummy is determined utilizing the Hertzian contact theory. The exchange of data between the dummy element and the flexible bridge is predicated on the principles of deformation compatibility and force equilibrium, as shown in Eqn 1 and 2. Among them, μ_{yy} is the dummy displacement, F_y is the horizontal constraint force, F_z is the vertical constraint force, Q is the wheel-rail vertical force, Y is the wheel-rail horizontal force, t is the running time, and s is the rail coordinate.

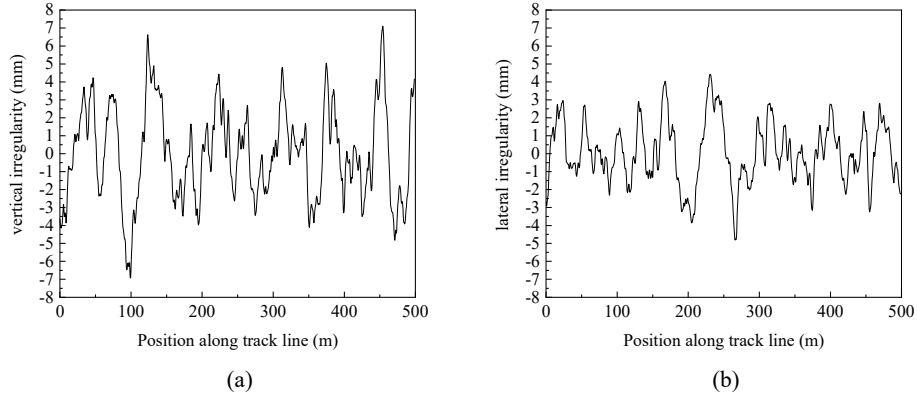


Fig. 3. Track irregularity simulation: (a) vertical irregularity; (b) lateral irregularity.

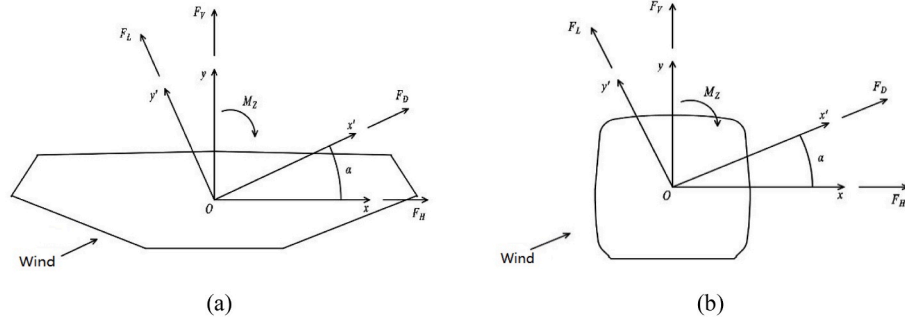


Fig. 4. Three-component force: (a) bridge section; (b) train vehicle section.

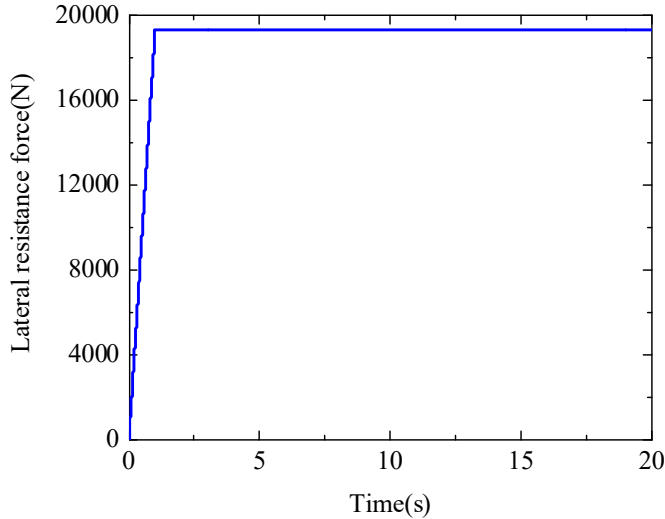


Fig. 5. Schematic diagram of the aerodynamic loading process.

Fig. 1 shows the flow chart of the wind-train-bridge-VIV coupled system

$$\begin{cases} Q(t) = F_z(t) \\ Y(t) = F_y(t) \end{cases} \quad (1)$$

$$\mu_{yy}(t) = \mu_b(t, s) \quad (2)$$

Among them, μ_{yy} is the displacement of the dummy, the lateral constraint force, F_z is the vertical constraint force, Q represents the vertical wheel-rail force, Y is the Lateral wheel-rail force, t is the running time, and s is the track coordinate.

2.1. Dynamical model of train

The dynamic model of the CRH2 train, featuring 42 degrees of freedom, is established within the SIMPACK (see Fig. 2). Force elements are utilized to simulate the suspension systems, and the entire train model is represented as a rigid body-spring-damping system. Moreover, the train model uses single-point geometric contact to define the geometric relationship of wheel-rail contact points.

For validating the precision of the multi-body dynamics system, the nominal force calculation method is used to obtain the maximum residual acceleration of the model. In this research, the maximum residual acceleration value for the model is only $3.62 \times 10^{-6} \text{ m/s}^2$, significantly lower than 0.01 m/s^2 . This demonstrates the reliability of the train model. The train set comprises eight units, including four motor cars and four trailers.

The low-interference Power Spectral Density (PSD) spectrum characteristic of the German high-speed railway is utilized to simulate track irregularities. The height irregularity, horizontal irregularity, and direction irregularity are taken as the input excitation (see Fig. 3).

2.2. Simulation of wind load

When constructing the wind-train-bridge coupled system under the impact of multiple excitations, it is crucial to account for the impact of wind forces. To simplify the wind load, the random natural wind is decomposed into fluctuating wind (V) and average wind (U). The three-component forces acting on the train and bridge are shown in Fig. 4. Within the body axis coordinate system, the unit static wind force acting on both the bridge and the train is expressed as follows:

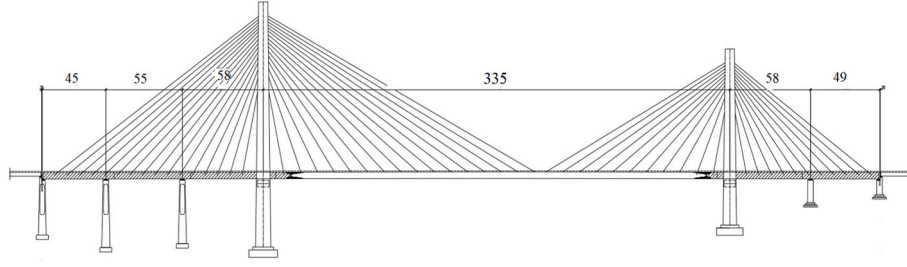


Fig. 6. Layout of the railway bridge (unit: m).

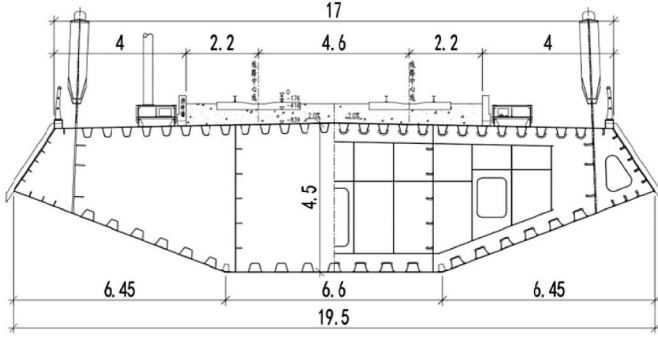


Fig. 7. The section of the girder (unit: m).

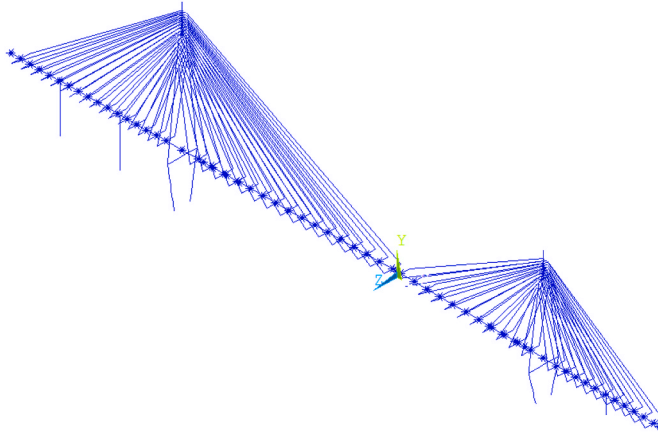


Fig. 8. Finite element model of cable-stayed bridge.

$$\begin{cases} D_0 = \frac{1}{2} \rho U^2 C_D(\alpha_0) H \\ L_0 = \frac{1}{2} \rho U^2 C_L(\alpha_0) B \\ M_0 = \frac{1}{2} \rho U^2 C_M(\alpha_0) B^2 \end{cases} \quad (3)$$

In the formula, L_0 and M_0 are respectively the static wind drag, lift, and moment per unit length in the body axis coordinate; ρ is the air density, generally taken as 1.225 kg/m^3 ; U is the average wind speed, the unit is m/s ; α_0 is the wind attack angle; H and B are the height and width of the bridge or train section; $C_D(\alpha_0)$, $C_L(\alpha_0)$ and $C_M(\alpha_0)$ are the three-component force coefficients of the body axis coordinate system at the wind attack angle of α_0 . The coefficients are obtained by the wind tunnel test.

Both non-uniform inflow distribution and turbulence reduce VIV response (Zhu et al., 2018). Considering both VIV and buffeting in the wind-train-bridge coupled system greatly increases complexity. Thus,

this study only performs wind-train-bridge-VIV coupled analysis under uniform flow, ignoring buffeting effect. Future research will further investigate the wind-train-bridge-VIV coupled analysis within a turbulent wind field.

The aerodynamic load is applied to the mass center of the vehicle and each node of the bridge. To weaken the sudden change of wind load during loading, the loading time of 1.0s is adopted in this paper, ignoring the unloading process (Han et al., 2020). Once loading stabilizes, the static wind force remains constant till calculations end. The loading process of wind load is shown in Fig. 5.

2.3. Simulation of VIV

This paper focuses on the wind-train-bridge coupled system, using the VIV displacement applied to the flexible girder, and investigates the response of the wind-train-bridge system under the VIV condition. Based on the wind tunnel test outcomes, the VIV is described as harmonic vibration. The VIV corresponding to the first-order vertical mode can be formulated as:

$$Y(t) = A\phi_1 \sin(2\pi ft + \theta) \quad (4)$$

where $Y(t)$ denotes the time-varying displacement of VIV, A represents the VIV amplitude obtained from the wind tunnel tests, f is the first-order vertical frequency, ϕ_1 is the mode vector corresponding to the f , θ represents the starting phase of VIV as the train commences its journey on the bridge. In the wind-train-bridge coupled model, the VIV is imposed on the corresponding girder nodes as excitations.

3. Engineering background

The subject of this study is a cable-stayed bridge distinguished by its varying tower heights. As shown in Fig. 6, the bridge has a length of 601.5 (45 + 55 + 58 + 335 + 58 + 49) m. The structure is a semi-floating system. The middle span of the bridge adopts the steel box girder, and the side span adopts the concrete girder. The section has a total width of 19.5 m, with the height at its center reaching 4.5 m (see Fig. 7).

Eigenvalue analysis was performed on the finite element model shown in Fig. 8 to determine the structure's frequencies and vibrational modes. Table 1 presents the computed dynamic properties of the bridge. In this table, L denotes the lateral direction, V signifies the vertical direction, S stands for symmetrical modes, and A represents the anti-symmetry.

Table 1
Dynamic characteristics of the bridge.

Mode	Frequency (Hz)	Mode shape
1	0.4295	V-S-1
2	0.4934	L-S-1
3	0.6244	V-A-1
5	0.8139	V-A-2
6	0.9566	V-S-2
10	1.2705	L-A-1
17	2.1045	T-S-1

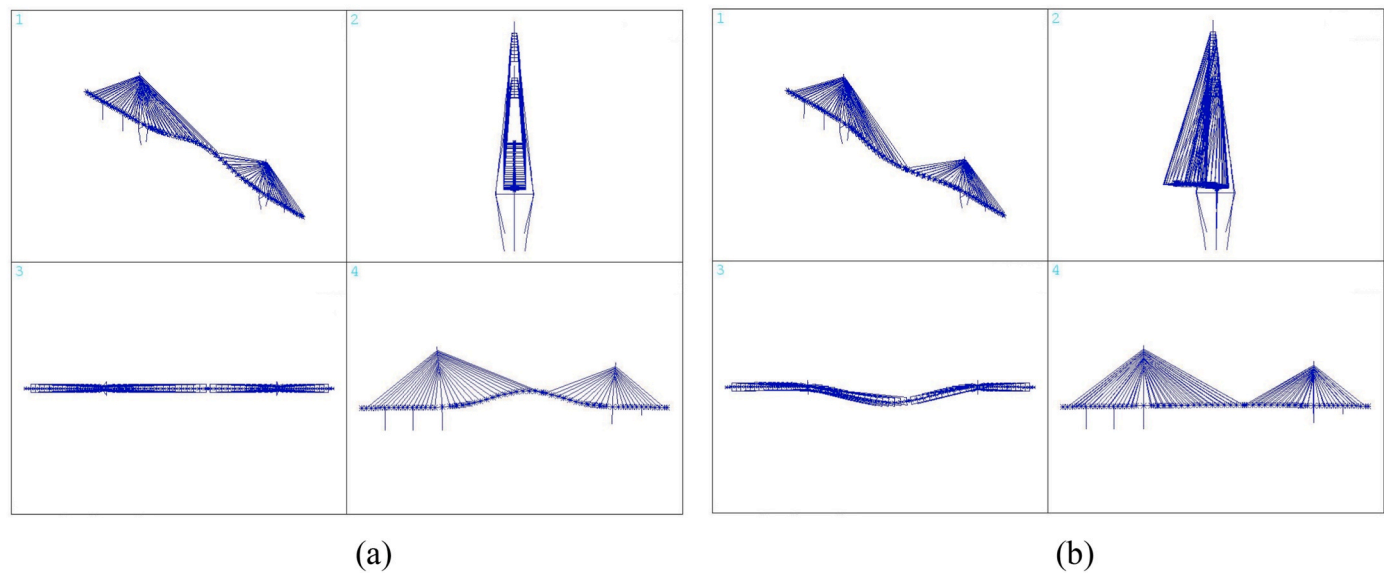


Fig. 9. Mode shapes: (a) First vertical bending mode; (b) First horizontal bending mode.



Fig. 10. Wind tunnel test for three-component force: (a) train model; (b) bridge model.

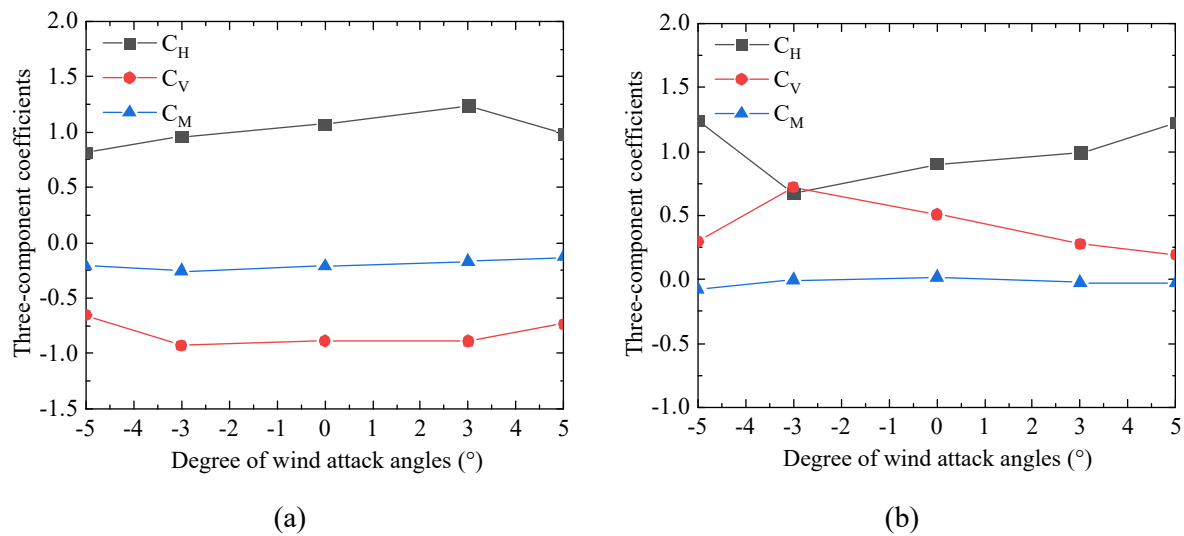


Fig. 11. Three-component coefficients: (a) bridge model; (b) train model.



Fig. 12. VIV wind tunnel test.

Table 2

The parameters of the scaled model according to the principle of similitude.

Parameter	Unit	Prototype	Modeling
Height (H)	m	4.5	0.15
Width (B)	m	19.5	0.65
Frequency (f_h)	Hz	0.430	2.688
Mass (m)	kg/m	35988	39.987
Damping ratio (ξ_h)	%	–	0.26

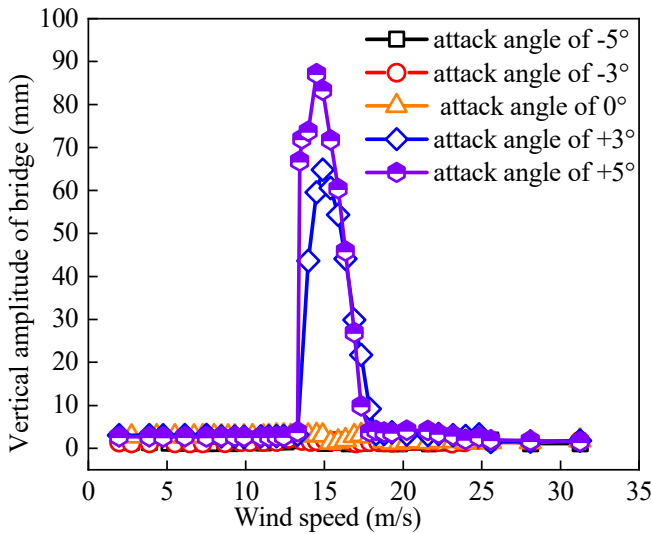


Fig. 13. Results of wind tunnel test of VIV under different wind attack angles.

The primary frequency of the bridge, measured at 0.4295 Hz, corresponds to the bridge's vertical bending mode (see Fig. 9). This suggests that the bridge's vertical bending stiffness is comparatively low. In addition, the first 10 vibration modes of the bridge do not include any torsional modes, so it can be considered that the bridge has good torsional stiffness.

4. Wind tunnel tests

4.1. Static wind loads coefficients

The experiment was conducted in the XNJD-1 wind tunnel. During the three-component force test, both sides of the model are directly mounted on the three-component force test balance. The orientation of the model is governed by the computer, ensuring precise manipulation of the attitude angle.

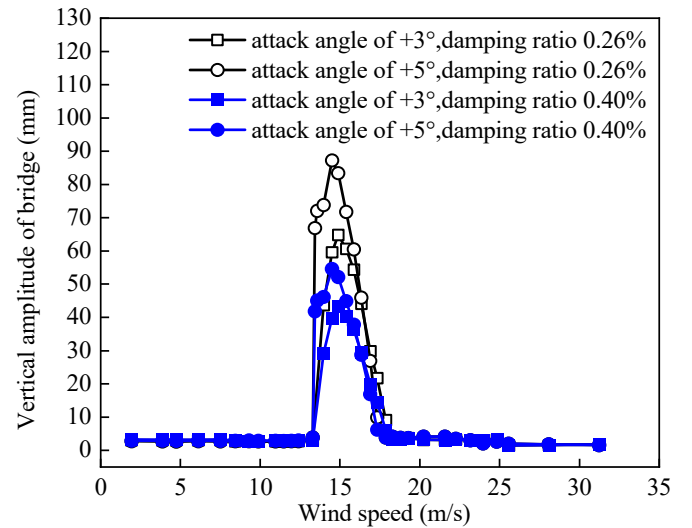


Fig. 14. Results of wind tunnel test of VIV under different damping ratios.

Considering the size of the model and the section size of the wind tunnel, this study employs a geometric scaling factor of 1:30 to make the model. The bridge model measures 2.095 m in length, 0.65 m in width, and 0.15 m in height, with a length-to-width ratio of 3.22. The train model is based on the central segment of the CRH2 train. The width of the model is 0.113 m and the height is 0.117 m. To mitigate the effects of flow field interference, endplates are positioned at the extremities of the section model (see Fig. 10). The experiment was conducted under uniform flow conditions. The three-component coefficients obtained by Eq. (3) are shown in Fig. 11.

4.2. VIV characteristics

The VIV test model is identical to the static three-component force test model, suspended by eight springs that allow for vertical movement and rotation around the model's center of gravity. The end plates are set to maintain the flow's two-dimensional characteristics around the model (See Fig. 12).

Furthermore, the model maintains geometric congruence with the actual bridge, the equivalence of the Strouhal number (St) should also be satisfied. The specific parameters are detailed in Table 2.

The wind tunnel test selected wind attack angles of -5° , -3° , 0° , $+3^\circ$, and $+5^\circ$. The outcomes revealed that the model experienced pronounced vertical VIV under conditions of a uniform flow field. The amplitudes and lock-in regions of the VIV are shown in Fig. 13. The girder exhibits a notable range of vertical VIV lock-in wind speeds under $+3^\circ$ and $+5^\circ$ wind attack angles, whereas no significant VIV phenomena are observed at 0° , -3° , and -5° wind attack angles. This indicates that the negative attack angle performance of the section is superior to that of positive attack angles. At wind attack angles of $+3^\circ$ and $+5^\circ$, the lock-in range for the VIV is between 13 and 18 m/s, with corresponding maximum VIV amplitudes of 64.8 mm and 87.2 mm, respectively.

Given that variations in the damping ratio can significantly influence the VIV response, this study also investigates the impact of diverse damping ratios on VIV behavior. As shown in Fig. 14 and Table 3, when the damping ratio increases from 0.26% to 0.40%, the maximum VIV amplitude at a $+3^\circ$ wind attack angle is reduced by 33.3%. Similarly, the maximum VIV amplitude at $+5^\circ$ attack angle is reduced by 37.5%. There is no significant change in the VIV interval of the two damping ratios.

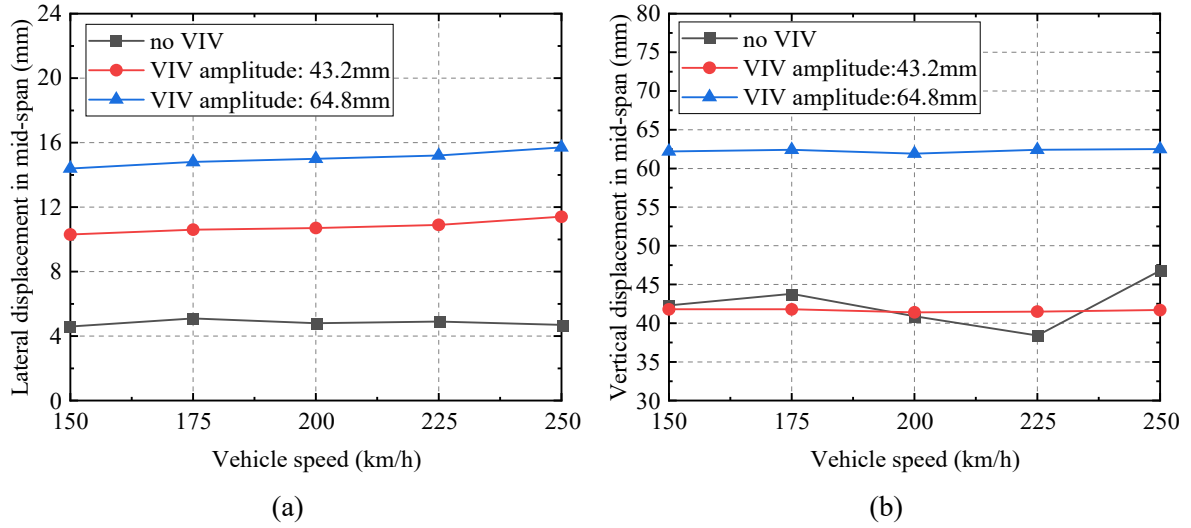


Fig. 15. The maximum displacement at the bridge's mid-span under various velocities: (a) lateral direction; (b) vertical direction.

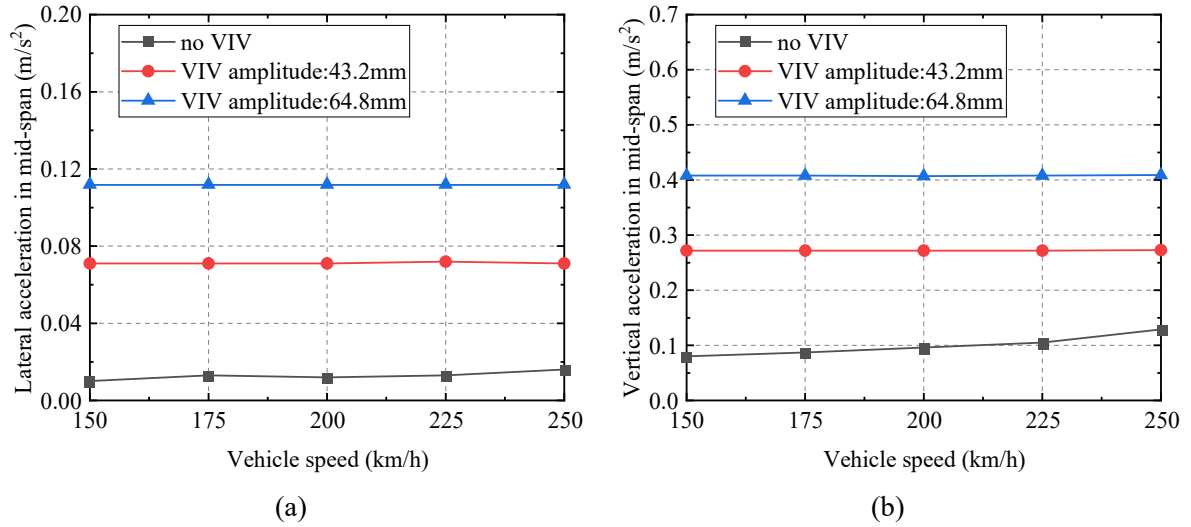


Fig. 16. The maximum acceleration at the bridge's mid-span under various velocities: (a) lateral direction; (b) vertical direction.

5. Responses of wind-train-bridge-VIV coupled system

5.1. Effects of running speeds

The wind-train-bridge-VIV coupled system was analyzed based on the scenario where the train passes through the bridge on the windward side, with a focus on the calculation results at a $+3^\circ$ wind attack angle. The maximum dynamic response of the bridge is shown in Figs. 15 and 16.

Examination of the charts reveals that the response of the bridge escalates in proportion to the VIV amplitude. The operation velocity of the train exerts minimal influence on the bridge's dynamic behavior under VIV conditions. Compared to the condition without VIV, at the speed of 250 km/h, the bridge's lateral displacement increased by 142.6 % and 234 % when the VIV amplitudes were 43.2 mm and 64.8 mm, respectively. Furthermore, under the condition of low VIV amplitudes, the vertical movement at the bridge's mid-span is less than that observed under the condition without VIV at certain speeds. At a velocity of 250 km/h and with a VIV amplitude of 43.2 mm, the bridge's vertical displacement is reduced by 10.9 % compared with the condition without

VIV. Moreover, when the train crosses the bridge under conditions of VIV, the peak vertical displacement at the bridge's mid-span is noted to be less than the amplitude of VIV. The reduction of the displacement is mainly because the train is regarded as a moving mass-spring-damping model in the system. This indicates that the train suppresses the vertical displacement of the bridge under VIV.

Figs. 17–19 illustrates the safety and comfort indices of the motor vehicle at various velocities. With the increase in speed, the safety and comfort indices of the train are on the rise. The vertical VIV exerts a minimal impact on the train's derailment coefficient, but it will significantly increase the wheel unloading rate. Both safety coefficients increase linearly with speed. When the speed is 250 km/h, the maximum derailment coefficient and wheel unloading rate are less than 0.6 under both amplitudes. As the VIV amplitude grows, the train's vertical acceleration increases significantly, while the lateral acceleration remains largely unaffected. At the speed of 250 km/h, the vertical acceleration of the train increased by 24.9 % and 51.0 % when the VIV amplitudes were 43.2 mm and 64.8 mm, respectively, compared to the condition without VIV. The variations in the train's various performance indices at different speeds, both with and without VIV, exhibit similar patterns.

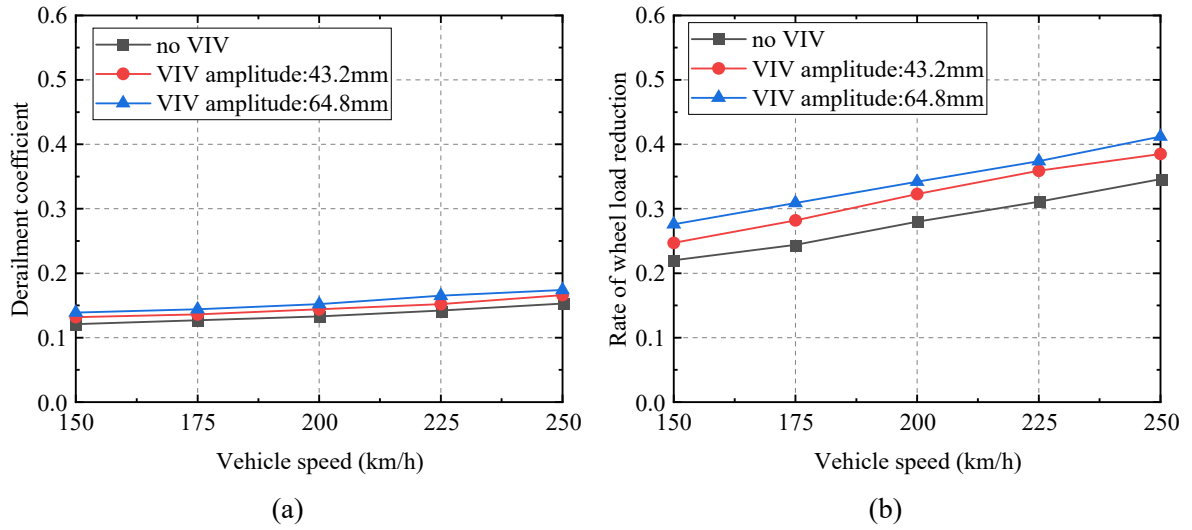


Fig. 17. The safety index of the motor vehicle at different speeds: (a) derailment coefficient; (b) wheel unloading rate.

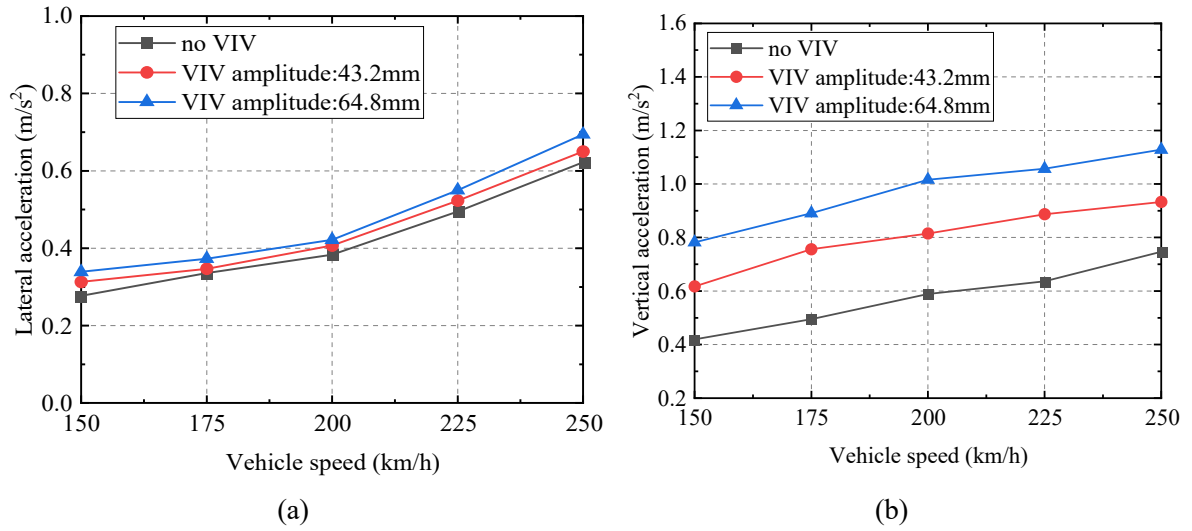


Fig. 18. Vibrating acceleration of the motor vehicle: (a) lateral direction; (b) vertical direction.

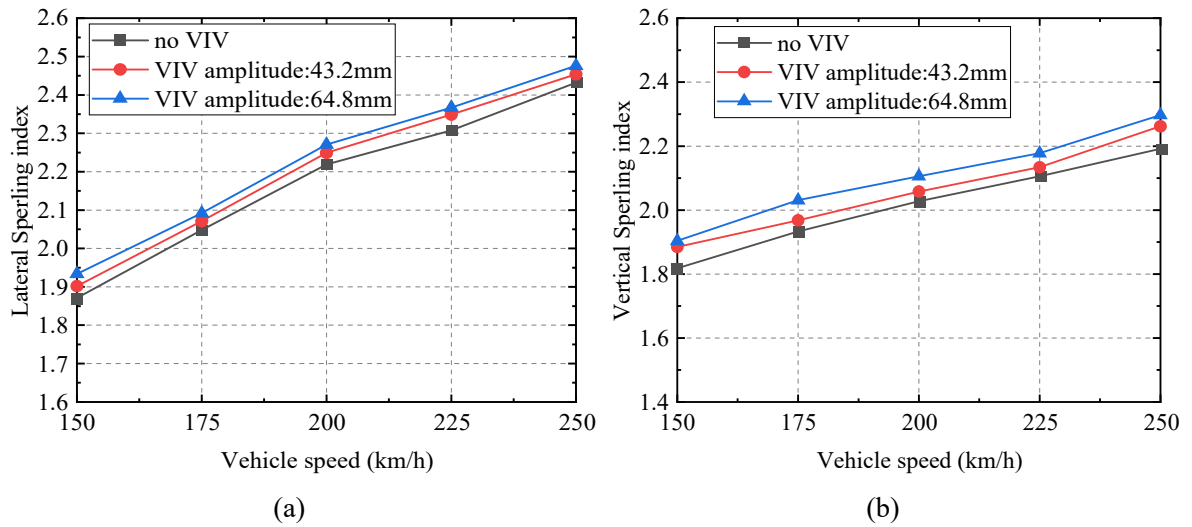


Fig. 19. Sperling index of the motor vehicle: (a) lateral direction; (b) vertical direction.

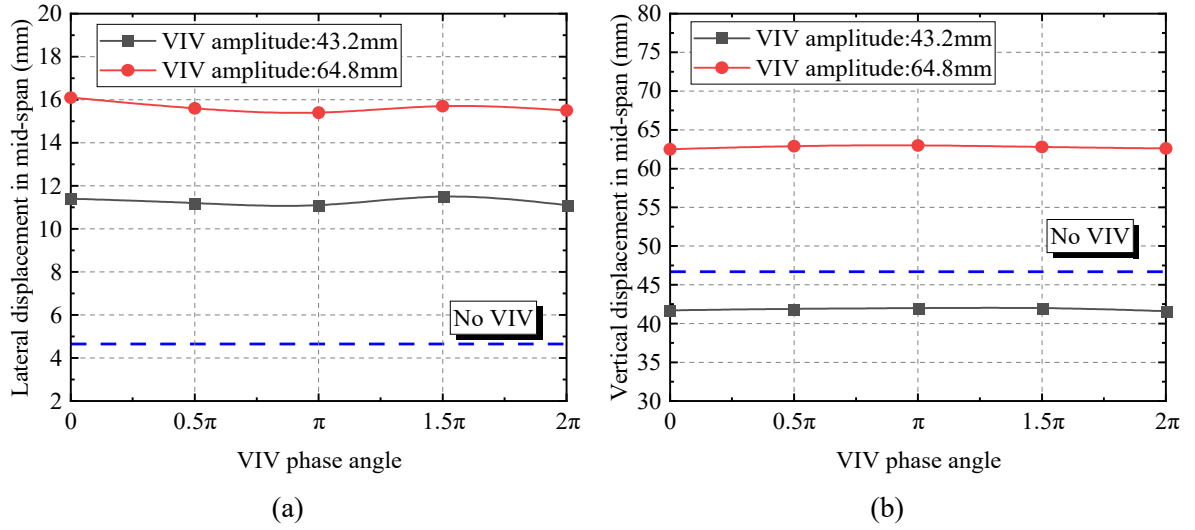


Fig. 20. The maximum displacement of the bridge mid-span at different phase angles: (a) lateral direction; (b) vertical direction.

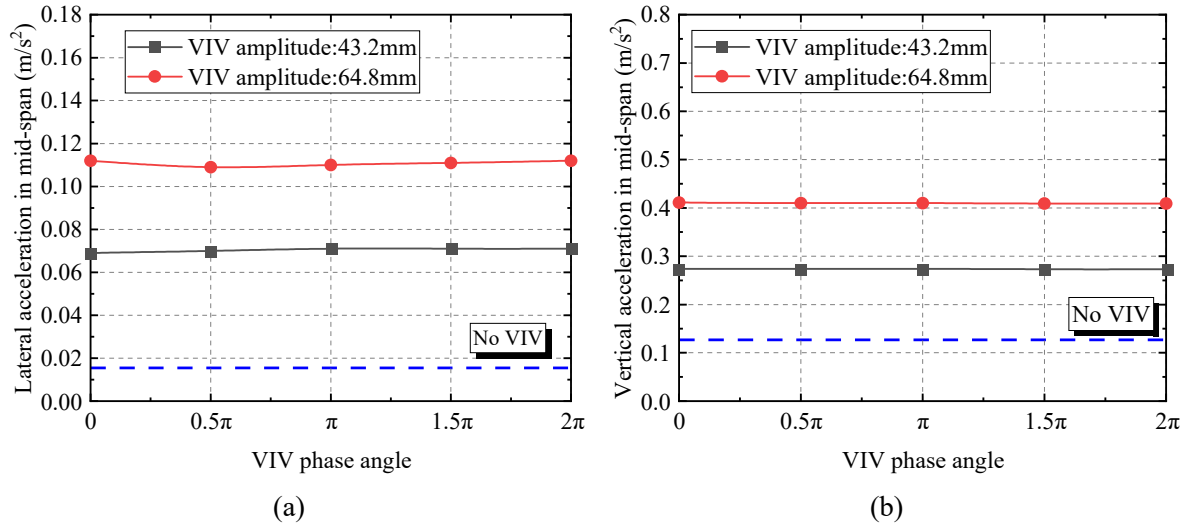


Fig. 21. The maximum acceleration of the bridge mid-span: (a) lateral; (b) vertical.

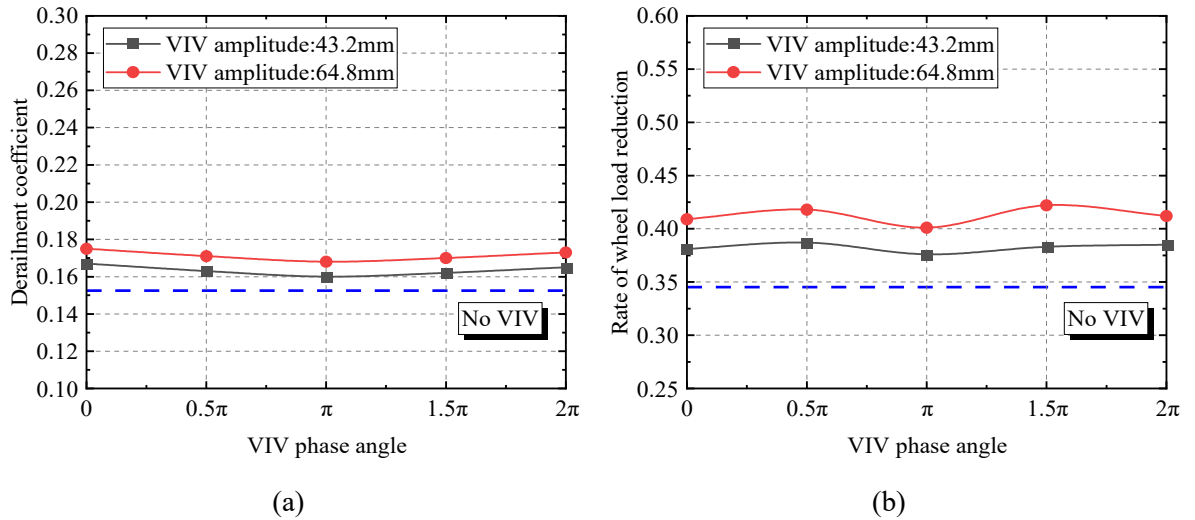


Fig. 22. The safety index of the motor vehicle: (a) derailment coefficient; (b) wheel unloading rate.

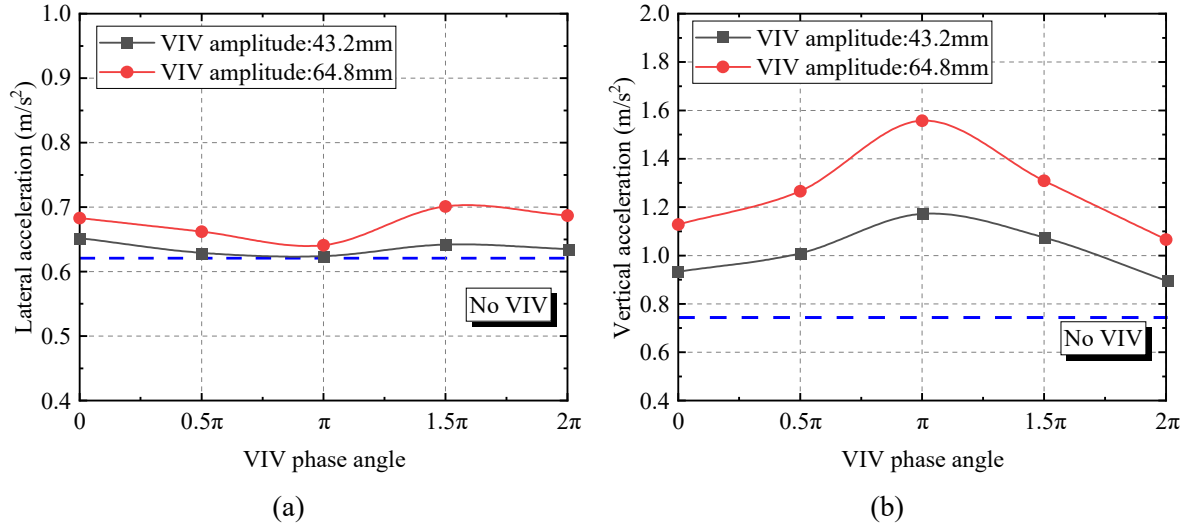


Fig. 23. Vibrating acceleration of the motor vehicle: (a) lateral; (b) vertical.

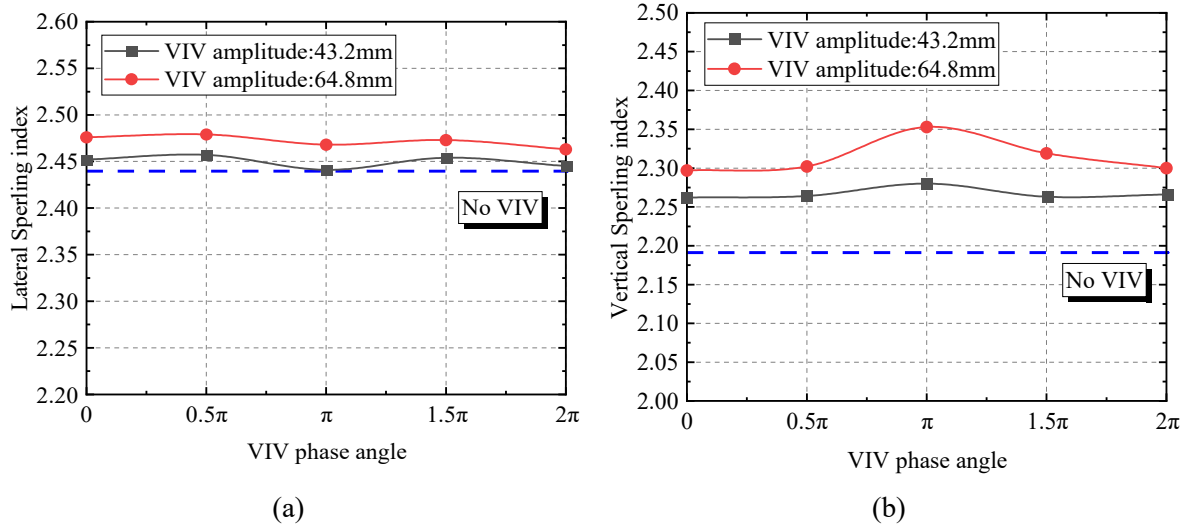


Fig. 24. Sperling index of the motor vehicle: (a) lateral direction; (b) vertical direction.

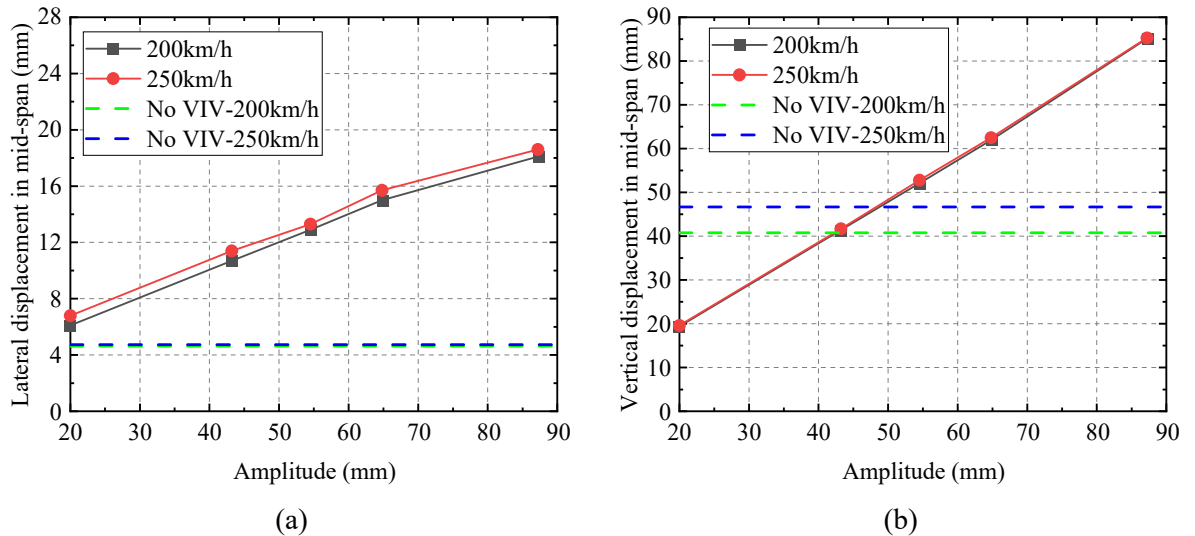


Fig. 25. The maximum displacement of the bridge mid-span: (a) lateral; (b) vertical.

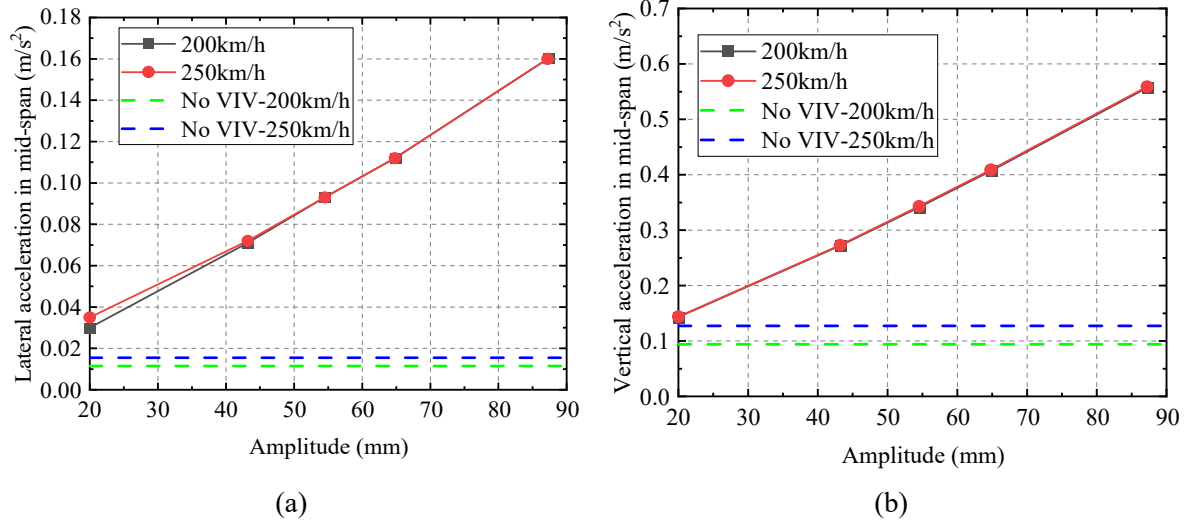


Fig. 26. The maximum acceleration of the bridge mid-span: (a) lateral; (b) vertical.

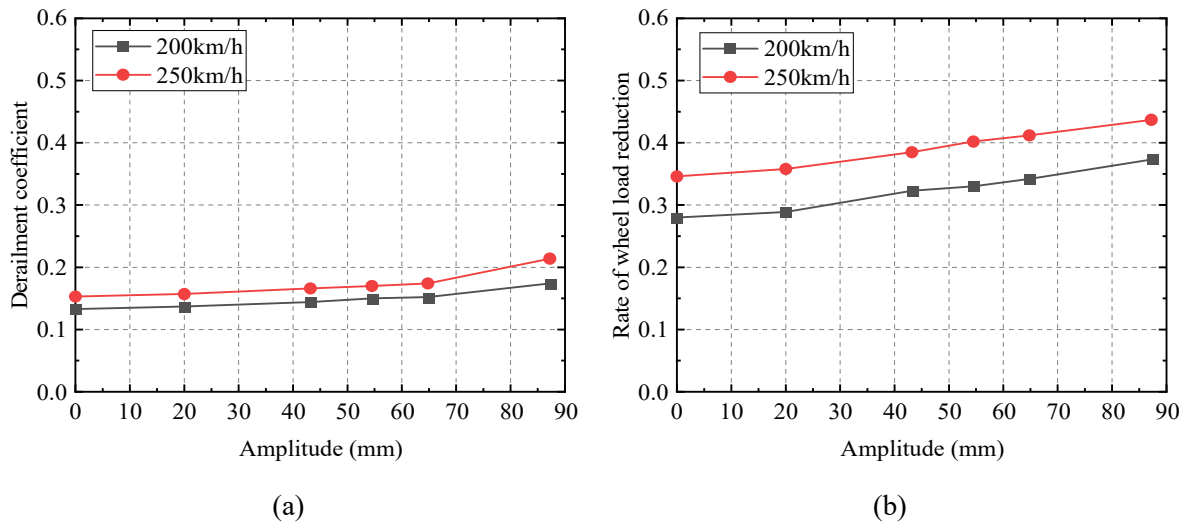


Fig. 27. The safety index of the motor vehicle: (a) derailment coefficient; (b) wheel unloading rate.

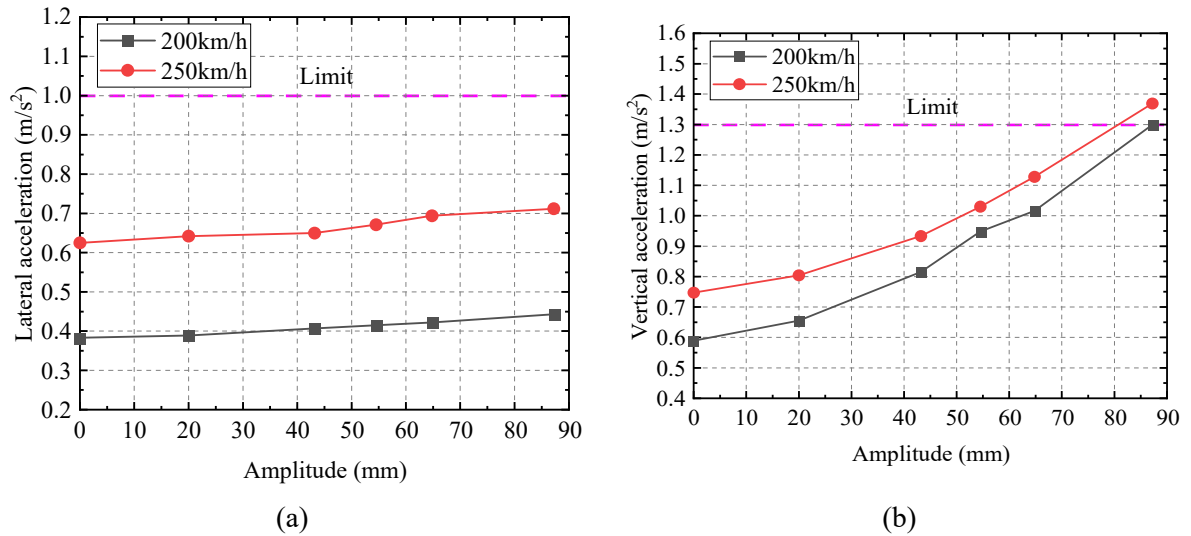


Fig. 28. Vibrating acceleration of the motor vehicle: (a) lateral; (b) vertical.

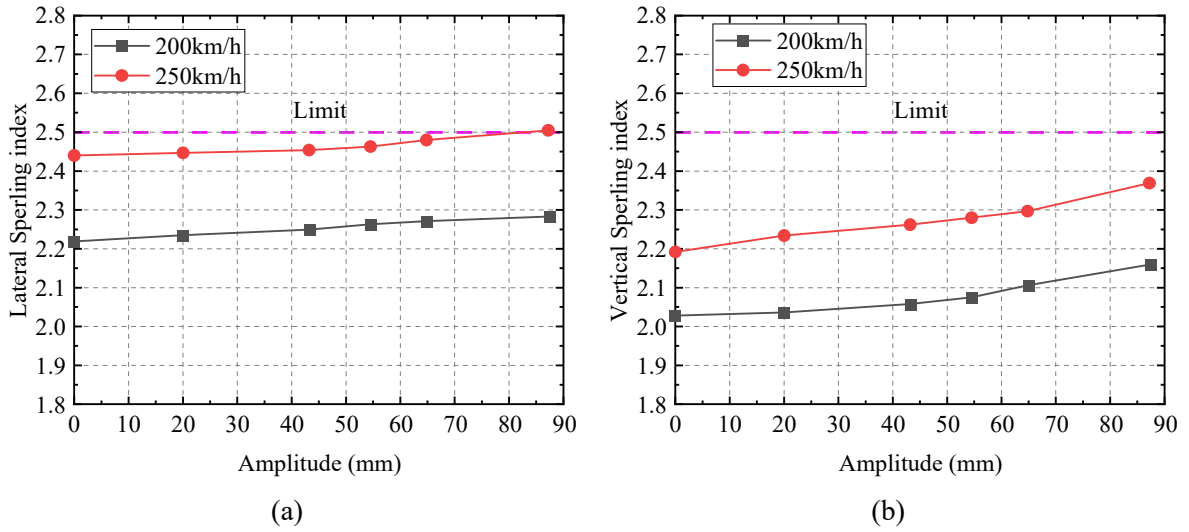


Fig. 29. Sperling index of the motor vehicle: (a) lateral; (b) vertical.

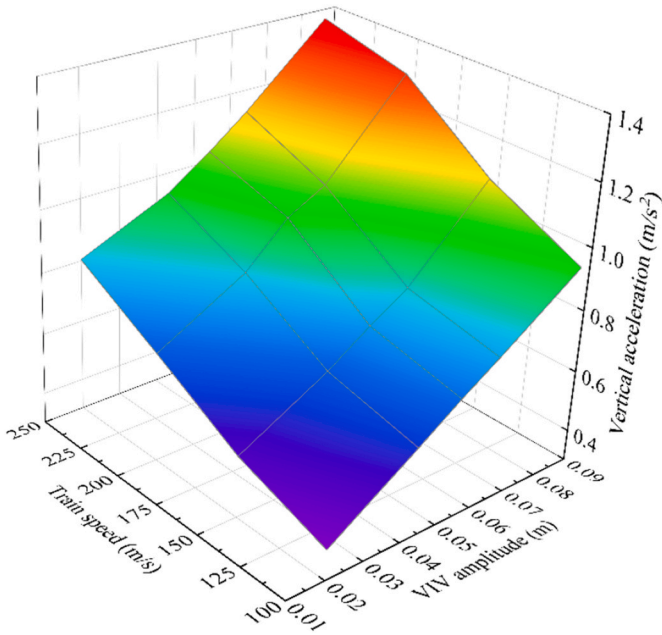


Fig. 30. Variation of train acceleration with respect to VIV amplitude at different speeds.

Consequently, speed remains a key factor affecting the lateral dynamic response of the train during VIV conditions.

5.2. Effects of phase angles

Calculations for the dynamic response of the wind-train-bridge-VIV coupled system are performed for a train traveling at 250 km/h under a $+3^\circ$ wind attack angle. The VIV phase angle ranges from 0 to 2π with increments of 0.5π .

From Figs. 20–21, it is evident that the bridge's dynamic response changes periodically with the phase angle in the range of 2π . Variations in the phase angle have a negligible effect on the bridge's dynamic response. At an amplitude of 43.2 mm for VIV, the vertical movement in the bridge mid-span is less than that without VIV, which is reduced by 11.1 %. The lateral displacement, along with both lateral and vertical accelerations of the bridge are greater than those without VIV.

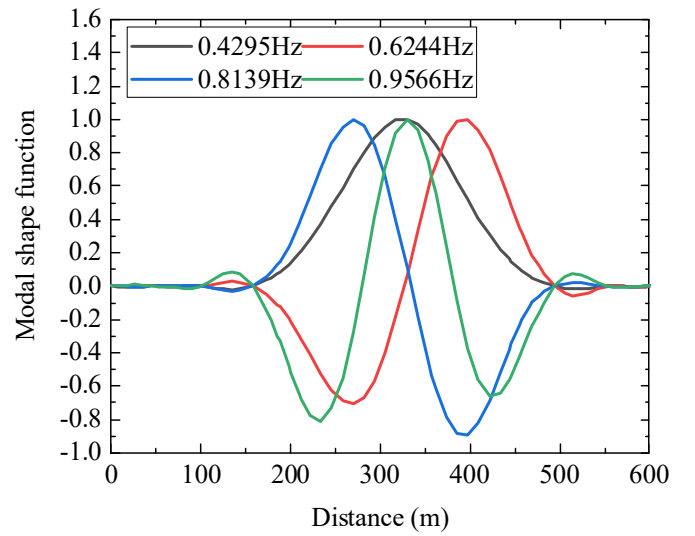


Fig. 31. The modal shape functions of the girder at different frequencies.

Specifically, the vertical acceleration experiences a substantial rise, amounting to an increment of 111.6 %. The amplitude of VIV remains the primary factor affecting the bridge's dynamic response.

As shown in Figs. 22–24, the vertical VIV exerts minimal influence on the derailment coefficient of the train. However, it is noteworthy that it amplifies the wheel unloading rate. Variations in the phase angle cause noticeable fluctuations in the wheel unloading rate, Sperling index, and vertical acceleration of the train. When the VIV amplitude is 64.8 mm, the vertical acceleration of the train can vary by up to 46.1 % at different phase angles; the wheel unloading rate can vary by a maximum of 5.2 % at different phase angles. The train's dynamic response exhibits a periodic fluctuation in correlation with the phase angle, yet this fluctuation does not synchronize with the changes in VIV excitation. This inconsistency is mainly attributed to the random irregularity on the track.

5.3. Effects of VIV amplitudes

The dynamic reactions of the wind-train-bridge-VIV coupled system are determined for the cases with train speeds of 200 km/h and 250 km/h under a $+3^\circ$ wind attack angle. The VIV amplitude is derived from wind tunnel testing. The maximum dynamic response indicators for the

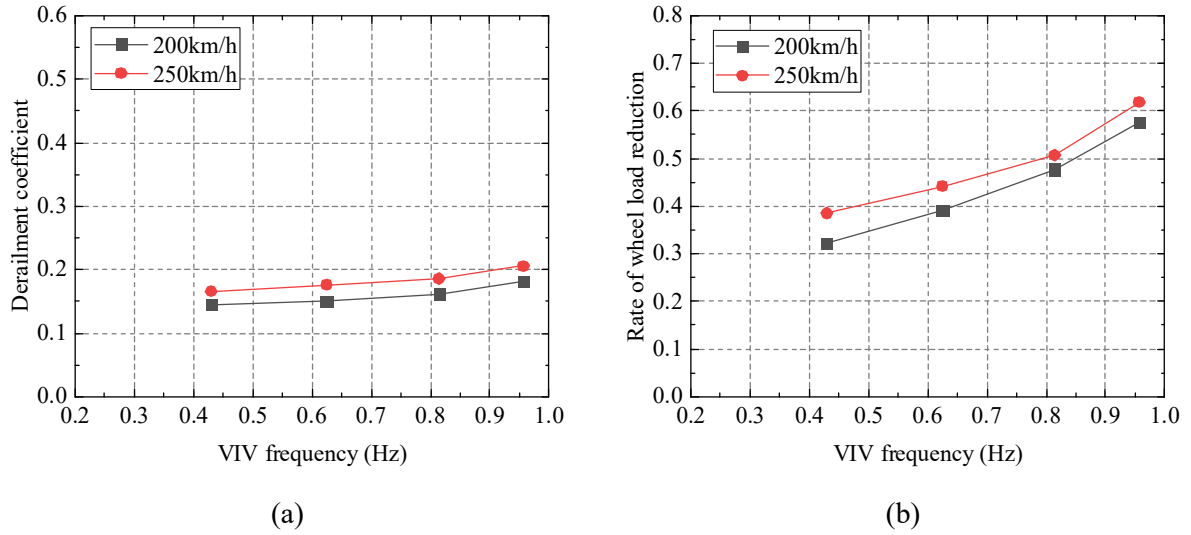


Fig. 32. The safety index of the motor vehicle: (a) derailment coefficient; (b) wheel unloading rate.

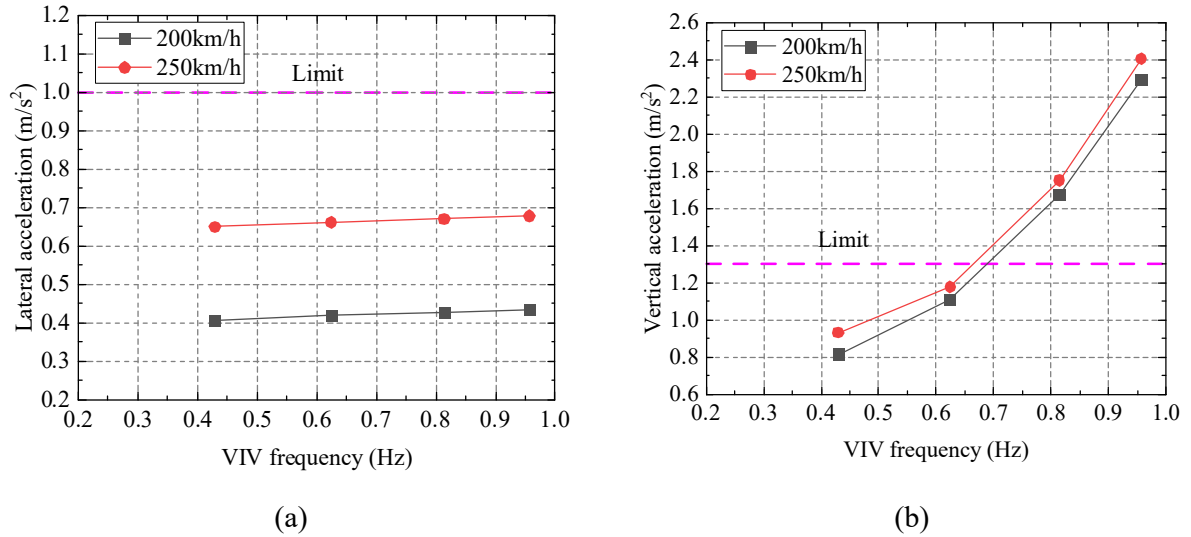


Fig. 33. Vibrating acceleration of the motor vehicle: (a) lateral; (b) vertical.

bridge are shown in Fig. 25–Fig. 26.

It is evident that as the VIV amplitude increases, so too does the dynamic response of the bridge. However, at a train speed of 200 km/h and with an amplitude below 43.2 mm, the vertical displacement at the bridge's mid-span is observed to be smaller compared to conditions without VIV. Similarly, at a velocity of 250 km/h, the vertical displacement at the mid-span remains smaller than in non-VIV conditions when the amplitude is below 46.8 mm. The findings suggest that the bridge's vertical dynamic response exhibits a linear increase in correlation with the rising amplitude of VIV.

As the VIV amplitude escalates, there is a notable increase in both the wheel unloading rate and the vertical acceleration experienced by the train. Conversely, the derailment coefficient and lateral comfort indices exhibit minimal increments. Furthermore, an augmentation in the VIV amplitude leads to a reduction in the disparity of vertical acceleration between two distinct train speeds. Upon reaching a speed of 250 km/h, the maximum lateral Sperling index for the motor vehicle attains its limit at an amplitude of 87.2 mm, while the maximum vertical acceleration for the motor vehicle surpasses the limit at the same amplitude. (see Figs. 27–29)

Under different train speeds, the train acceleration varies with the

VIV amplitude as shown in Fig. 30.

As shown in Fig. 30, the vertical acceleration of the train has a nonlinear relationship with train speed and VIV amplitude. As the train velocity decreases to near-zero values, the relationship between VIV amplitude and vertical acceleration of the train system transitions into a linear regime. Conversely, high train speeds significantly amplify the nonlinearity of train vertical acceleration under VIV. Moreover, the combination of high speed and high VIV amplitude leads to a sharp increase in vertical acceleration.

5.4. Effects of VIV frequency

As shown in the previous analysis, the influence of the train on the dynamic response of the bridge during VIV is negligible. Instead, VIV itself dominates the dynamic behavior of the bridge during the train-crossing process. Accordingly, this section focuses solely on the effects of different VIV frequencies and modes on the dynamic response of the train as it crosses the bridge.

Based on the cases outlined in Section 4.3, VIV frequencies and modal shapes are varied while keeping steady wind loading constant. The VIV amplitude is set to 43.2 mm in the calculations. The modal

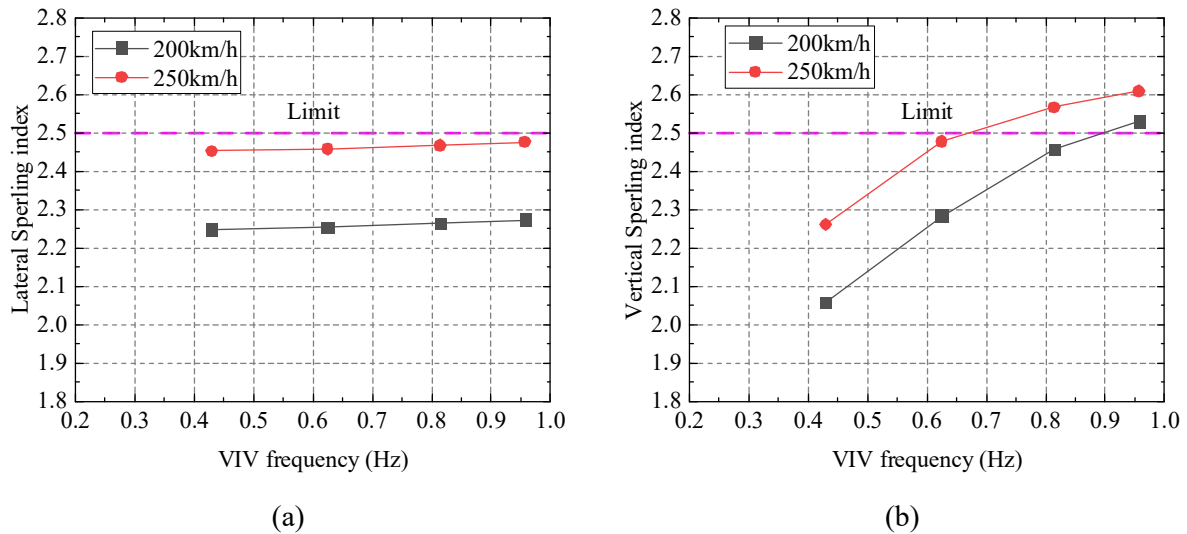


Fig. 34. Sperling index of the motor vehicle: (a) lateral; (b) vertical.

Table 3

Summary of wind tunnel test results of vertical VIV.

Attack angle	Damping ratio (%)	lock-in region (m/s)	Maximum amplitude (mm)	Wind speed at maximum amplitude (m/s)
+3°	0.26	13.9–17.8	64.8	14.9
	0.40	13.4–17.3	43.2	14.9
+5°	0.26	13.9–17.8	87.2	14.5
	0.40	13.4–17.3	54.5	14.5

Table 4

Evaluation criteria for driving comfort of the train.

Sperling index	Excellent: ≤ 2.5 ; good: 2.5–2.75; fair: 2.75–3.0		
maximum acceleration	medium speed ≤ 200 km/h	Vertical: $\leq 0.25g$; horizontal: $\leq 0.20g$;	
	medium speed ≥ 200 km/h	Vertical: $\leq 0.13g$; horizontal: $\leq 0.10g$;	

Table 5

Fitting curve parameters of vertical maximum acceleration.

Vehicle type	Speed (km/h)	B	C	D	R2
Motor vehicle	200	0.553	4.051	51.363	0.997
	250	0.734	2.247	57.970	0.999
Trailer vehicle	200	0.680	−1.271	162.520	0.999
	250	0.866	−2.943	182.976	0.991

shapes and frequencies used in the analysis are presented in Fig. 31.

As shown in Fig. 32, the derailment coefficient exhibits minimal variation with increasing frequency, whereas the wheel unloading rate increases significantly with rising frequency. At a VIV amplitude of 43.2 mm, derailment risk remains low across all tested VIV frequencies. However, at high VIV frequencies and amplitudes, the wheel unloading rate may exceed permissible limits, potentially compromising operational safety.

Figs. 33 and 34 show the variation trends of lateral and vertical accelerations and Sperling indices with VIV frequency at different train speeds. The lateral acceleration of the train is largely unaffected by changes in VIV frequency. In contrast, the vertical acceleration increases significantly with rising VIV frequency. At a frequency of 0.8139 Hz and a VIV amplitude of 43.2 mm, the train's vertical acceleration surpasses the allowable limit at both tested speeds, thereby degrading ride comfort. For the lateral Sperling index, both 200 km/h and 250 km/h exhibit

stable trends with increasing VIV frequency. The 250 km/h value is consistently higher than 200 km/h, reflecting a speed-dependent increase, yet frequency-induced variation remains negligible. In the vertical direction, the Sperling index for both speeds rise with VIV frequency, indicating that increased speed and frequency significantly degrade vertical ride quality, leading to elevated Sperling indices. Notably, the rate of increase in the vertical Sperling index diminishes as VIV frequency continues to rise.

6. VIV amplitude limit based on driving comfort

6.1. Evaluation criteria for driving comfort

The train's comfort index includes the Sperling index and the maximum acceleration. The limits for these indices are detailed in the specifications of China (GB/T 5599:1985, 1985) and the literature by Deng et al. (2019), as presented in Table 4.

Compared with the lateral and vertical comfort indices of the train, it is found that vertical VIV exerts the most significant impact on the train's vertical acceleration. At an amplitude of 87.2 mm for VIV, the train exceeds the allowable acceleration threshold at both speeds. Therefore, the VIV amplitude limit for railway bridges should be calculated based on the train's maximum acceleration.

6.2. VIV amplitude limit value

The VIV amplitude limit for the first vertical mode under VIV is determined based on the acceleration threshold. A nonlinear fit is applied to the train's vertical acceleration data to establish the relationship between the VIV amplitude and the maximum vertical acceleration of the train. The fitting function is:

$$y = Bx^2 + Cx + D \quad (5)$$

The fitting has converged, and the relevant parameters are presented in Table 5.

The fitting results in Fig. 35 indicate that at 250 km/h, the motor's vertical acceleration reaches its threshold when exposed to VIV with an amplitude of 81.3 mm. When the velocity is 200 km/h, the maximum vertical acceleration of the motor vehicle reaches the limit value at the amplitude of 87.4 mm. At velocities of 200 km/h and 250 km/h for the trailer, the VIV amplitudes corresponding to the vertical acceleration threshold are 65.8 mm and 57.3 mm, respectively. Exceeding these amplitudes is detrimental to the vertical comfort of the train traversing

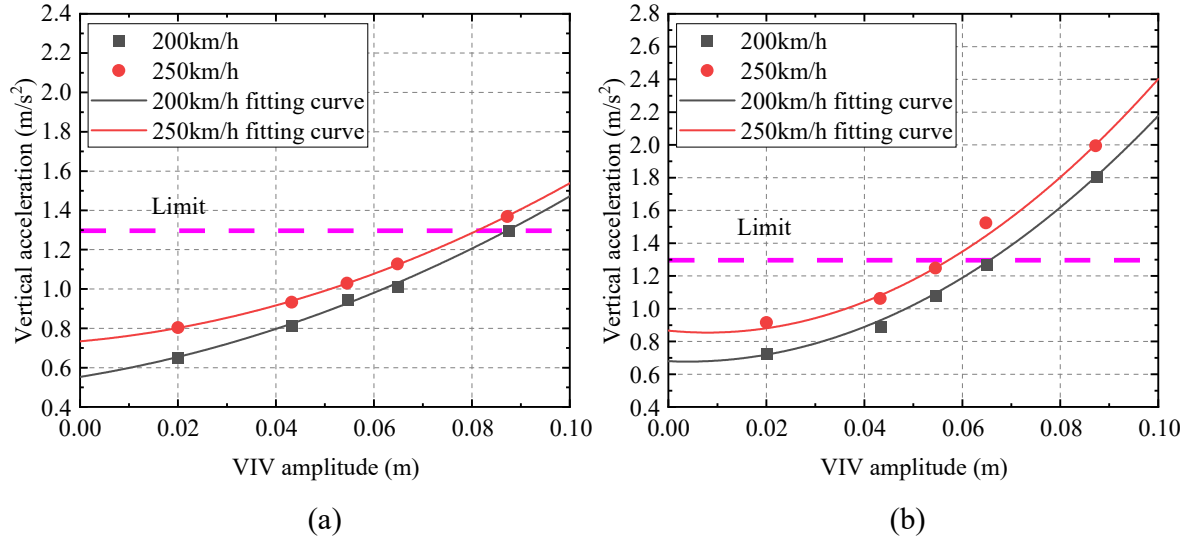


Fig. 35. The relationship between the maximum vertical acceleration of the train and VIV amplitude: (a) motor vehicle; (b) trailer vehicle.

Table 6

Fitting curve parameters of vertical maximum acceleration.

Speed (km/h)	E	F	R2
200	6.165	0.011	0.998
250	6.183	0.012	0.998

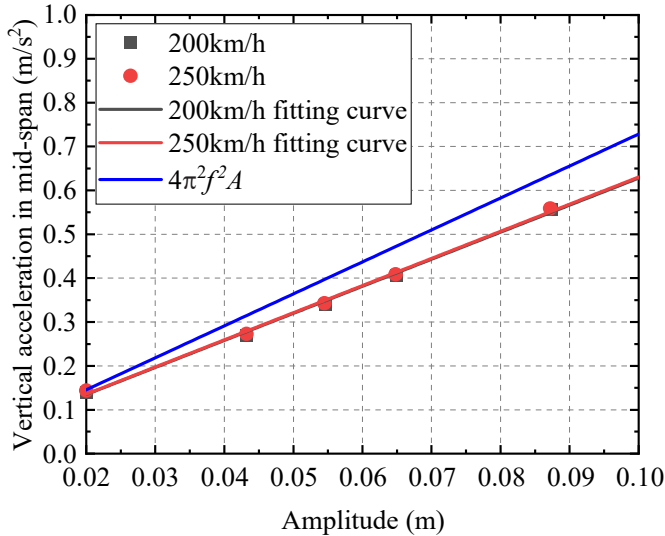


Fig. 36. The relationship between the maximum vertical acceleration of the bridge and VIV amplitude.

the bridge.

Since the acceleration at the bridge's mid-span correlates nearly linearly with the VIV amplitude, this paper performs a linear fit of the mid-span acceleration at different amplitudes. The fitting function is:

$$y = Ex + F \quad (6)$$

The fitting has converged, and the relevant parameters are presented in Table 6.

As can be seen from Eq. (4), the acceleration of the VIV of the bridge can be expressed as:

$$Y''(t) = (2\pi f)^2 A \phi_1 \sin(2\pi f t + \theta) \quad (7)$$

Therefore, the maximum acceleration generated by VIV in bridges is

$$Y''(t)_{\max} = (2\pi f)^2 A.$$

As shown in Fig. 36, when a train traverses the bridge, the maximum bridge acceleration under VIV is reduced. When the VIV amplitude reaches 87.2 mm, the mid-span bridge acceleration at a train speed of 250 km/h decreases by 12 % compared to the bridge with no train. As the VIV amplitude increases, the difference in maximum acceleration between the bridge with and without the train gradually widens.

Given that the impact of VIV on the trailer is more pronounced than on the motor vehicle, the subsequent analysis of the VIV amplitude limits presented in this paper focuses on the trailer. Since the operation speed has almost no impact on the bridge's maximum acceleration under VIV, the results with a speed of 250 km/h are used when analyzing the limit value of the VIV amplitude. (See Fig. 37)

6.3. Comparison of different VIV amplitude limits

In the “Wind-resistant Design Specification for Highway Bridges” (JTG/T 3360-01-2018, 2018), the vertical VIV amplitude limit formula of the bridge with a span of less than 200m is $\frac{0.04}{f_v}$. In its explanatory notes, if the amplitude limit value calculation formula is based on acceleration as the evaluation criterion for VIV, then the formula is $\frac{0.025}{f_v^2}$. In which f_v is the vertical vibration frequency of the bridge. According to the “Wind-resistant Design Specification for Railway Bridges” (Q/CR 9162-2023, 2023), the formula for the amplitude limit of VIV in vertical bending mode is:

$$\frac{0.01}{f_v^2 + \gamma_n \left(\frac{v}{L}\right)^2} \quad (8)$$

in which f_v is the vertical vibration frequency of the bridge; γ_n represents the vertical VIV mode coefficient, where the first-order symmetric mode takes 0.5, and the first-order anti-symmetric mode takes 1.0; v is the traveling speed; L is the main span length.

The British standard (CD 363, 2020) uses the dynamic sensitivity parameter K_D to assess VIV's impact on bridges' performance.

$$K_D = y_{\max} f_v^2 = \frac{1000}{4\pi^2} \frac{4\pi^2 y_{\max} f_v^2}{1000} = 25.33 A_D \quad (9)$$

in which f_v is the vertical vibration frequency of the bridge; $y_{\max}$

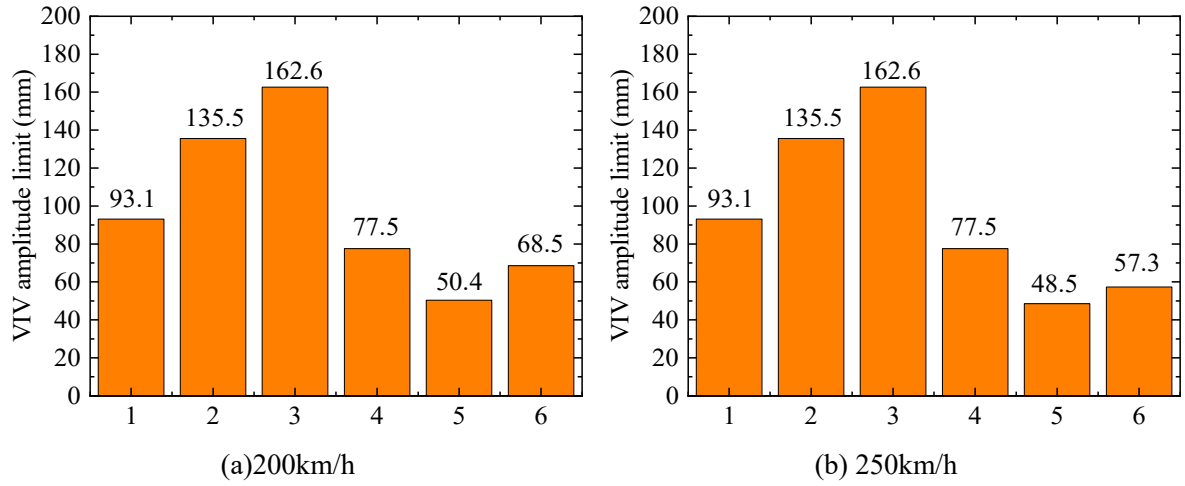


Fig. 37. Comparison of vertical VIV amplitude limits: (a) 200 km/h; (b) 200 km/h.

Table 7
Comparison of calculation results of VIV amplitude limit.

Method	Calculation method of limit value	VIV amplitude limit (mm)	
		200 km/h	250 km/h
1	0.04	93.1	
2	$\frac{f_v}{f_v^2}$	135.5	
3	$K_D = 30\text{mm/s}^2$	162.6	
4	$0.05g$	77.5	
5	0.01	50.4	48.5
6	$f_v^2 + \gamma_n \left(\frac{v}{L}\right)^2$		
	Calculation results	68.5	57.3

represents the vertical VIV amplitude; A_D represents the amplitude of acceleration for the bridge. Based on the correlation between the dynamic sensitivity parameter K_D and human comfort levels, the acceleration limit for VIV is reached when $K_D = 30\text{mm/s}^2$ ($A_D = 1.17\text{m/s}^2$).

According to the AASHTO LRFD Bridge Design Specifications (AASHTO LRFD-2020, 2020) for design wind speeds below 13.41 m/s, the amplitude of the vertical acceleration of the girder due to VIV should not exceed $0.05g$ (0.491m/s^2); when the design wind speed is $13.41\text{ m/s} < U \leq 22.35\text{ m/s}$, it should not exceed $0.1g$ (0.980m/s^2). Based on the fitted results of the maximum acceleration at the bridge's mid-span, the amplitude limit of VIV can be inferred.

According to the formula provided in the specifications and the previous calculation results, the amplitude limit for VIV of cable-stayed bridges can be obtained (See Table 7).

The conclusions can be drawn from the aforementioned charts and graphs:

- (1). The VIV amplitudes derived from the "Wind-resistant Design Specification for Highway Bridges" are larger than that obtained from the simulation calculation. At a speed of 250 km/h, the limits of VIV obtained by Method 1 and Method 2 are 62.5 % and 136.5 % higher than the simulation result. Moreover, the influence of operation speed is not taken into consideration. Therefore, the VIV limit stipulated in highway standards is potentially hazardous for large-span railway bridges, and thus, it is not fully applicable to railway bridges.
- (2). The relevant specifications in the UK and the US limit the amplitude of VIV based on the bridge's maximum acceleration. Both limits are higher than the simulation result. At a speed of 250 km/h, the UK standard is 183.8 % higher than the simulation

result, while the US standard is 35.3 % higher, which is very unfavorable for the comfort of trains traveling on the bridge.

- (3). The simulation results match well with the calculations from the "Wind Resistance Design Specifications for Railway Bridges." At the speed of 200 km/h, the specification is 26.4 % lower than the simulation result; at 250 km/h, it is 15.4 % lower. Furthermore, the VIV limits derived from calculations from "Wind-resistant Design Specification for Railway Bridges" are lower than those obtained from simulation calculations. This suggests that the VIV amplitude limits in "Wind-resistant Design Specification for Railway Bridges" are more conservative.
- (4). Upon decreasing the speed from 250 km/h to 200 km/h, the VIV amplitude limit calculated by "Wind-resistant Design Specification for Railway Bridges" increases by 3.8 %, whereas the limit obtained from simulation calculations increases by 16.4 %. This suggests that the effect of reducing train speed on the amplitude limit of vertical VIV is limited.

7. Conclusions

This study examines the VIV performance of a cable-stayed railway bridge through wind tunnel tests. Within the wind-train-bridge coupled system framework, it assesses the system's dynamic response under VIV. The railway bridge's permissible VIV amplitude threshold is computed using the train's comfort indices and the bridge's maximum acceleration. Moreover, the VIV amplitude limits under various specifications are examined and compared with the coupled system's results. The main conclusions are summarized below:

- (1). The variation of the train's comfort and safety indices with the operation speed is similar to that without VIV. Train speed has minimal impact on bridge dynamics, and reducing speed from 250 km/h to 200 km/h only increases the VIV amplitude limit by 16.4 %.
- (2). VIV phase angle introduces periodic variations in train dynamic responses, with vertical acceleration and wheel unloading rate fluctuating by up to 46.1 % and 5.2 %, respectively, at an amplitude of 64.8 mm. Therefore, the impact of the VIV phase angle should be considered when determining the amplitude limit for VIV.
- (3). The vertical VIV amplitude substantially affects the train and bridge's vertical dynamic response, yet barely impacts the lateral one. It has a minimal effect on the train's derailment coefficient, but will greatly increase the wheel unloading rate. Results showed that at 250 km/h, compared with no VIV, the train's

vertical acceleration rose by 24.9 % with a 43.2 mm VIV amplitude, and by 51.0 % with a 64.8 mm amplitude.

- (4) While the derailment coefficient remains relatively stable with increasing VIV frequency, the wheel unloading rate rises significantly, particularly at high VIV frequencies and amplitudes, where it may pose operational risks. Additionally, lateral acceleration and lateral Sperling indices show minimal sensitivity to VIV frequency, whereas vertical acceleration and vertical Sperling indices increase with rising VIV frequency, indicating degraded vertical ride quality. Notably, the rate of increase in vertical Sperling indices reduces as VIV frequency continues to rise.
- (5) For the 335-m span cable-stayed bridge, evaluated by train comfort, the vertical VIV amplitude thresholds at 200 km/h and 250 km/h are 65.8 mm and 57.3 mm respectively. Exceeding these limits will significantly compromise the vertical ride comfort of trains crossing the bridge.
- (6) Existing design codes (e.g., AASHTO LRFD-2020, Highway Bridge Wind Resistance Design Specifications) overestimate allowable VIV amplitudes when using bridge acceleration as the sole criterion. Using bridge maximum acceleration under VIV as the evaluation criterion fails to account for operational speed's impact on VIV limits, making these formulas unsuitable for railway bridge design. Instead, the "Wind-resistant Design Specification for Railway Bridges" offers amplitude limits better aligned with simulations and more appropriate for such designs.
- (7) The train's acceleration limits specified in the regulations may not fully reflect the comfort experienced by passengers. Future work will propose a flexible dynamic model of seated passengers to assess VIV's impact on human comfort and enhance the multi-objective evaluation system.

CRedit authorship contribution statement

Tianyu Chen: Writing – original draft. **Cunming Ma:** Project administration. **Qingsong Duan:** Writing – review & editing. **Kai Cheng:** Writing – review & editing, Data curation. **Cheng Pei:** Investigation.

Declaration of competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The financial support from the Key Research and Development Program of the People's Republic of China (Project No. 2022YFC3005300) and the Natural Science Foundation of China (Grant

No. 52078438) is gratefully acknowledged.

Data availability

Data will be made available on request.

References

- AASHTO LRFD-2020, Bridge Design Specifications.
 CD 363, Design Rules for Aerodynamic Effects on Bridges.
 Chen, X., Ma, C., Xian, R., Liu, T., 2022. Effects of gap configuration on vortex-induced vibration of twin-box girder and countermeasure study. *J. Wind Eng. Ind. Aerod.* 231, 105232.
 Deng, E., Yang, W., Lei, M., Zhu, M., Zhang, Z., Ping, P., 2019. Aerodynamic loads and traffic safety of high-speed trains when passing through two windproof facilities under crosswind: a comparative study. *Eng. Struct.* 188, 320–339.
 Duan, Q., Ma, C., Li, Q., 2023. Vortex-induced vibration performance and mechanism of long-span railway bridge streamlined box girder. *Adv. Struct. Eng.* 26 (13), 2506–2519.
 Fu, Y., Wu, F., Chai, C., Cui, W., Li, Y., Zhao, L., 2025. Assessment of driving safety and comfort during vortex-induced vibrations in a long-span bridge considering wind-vehicle-bridge interactions. *J. Wind Eng. Ind. Aerod.* 257, 106007.
 Railway vehicles—specification for evaluation the dynamic performance and accreditation test, GB/T 5599:1985, 1985.
 Han, Y., Liu, Y., Hu, P., Cai, C.S., Xu, G., Huang, J., 2020. Effect of unsteady aerodynamic loads on driving safety and comfort of trains running on bridges. *Adv. Struct. Eng.* 23 (13), 2898–2910.
 Ji, Z., Yang, G., Liu, Y., Jiang, Q., 2018. Analysis of vertical vibration characteristics of the vehicle-flexible track coupling system under wind load and track irregularity. *Proc. Inst. Mech. Eng. F J. Rail Rapid Transit* 232 (10), 2444–2455.
 Wind-resistant design specification for highway bridges, JTG/T 3360-01-2018, 2018.
 Li, Y., Xu, X., Zhou, Y., Cai, C.S., Qin, J., 2018. An interactive method for the analysis of the simulation of vehicle-bridge coupling vibration using ANSYS and SIMPACK. *Proc. Inst. Mech. Eng. F J. Rail Rapid Transit* 232 (3), 663–679.
 Li, H.Y., Xu, Y.L., Wang, B., Zhu, L.D., Meng, X.L., Zhang, G.Q., 2024. Establishment and application of a digital twin for vortex-induced vibration of a bridge deck section. *Eng. Appl. Comput. Fluid Mech.* 18 (1), 2297032.
 Noguchi, K., Ito, Y., Yagi, T., 2020. Numerical evaluation of vortex-induced vibration amplitude of a box girder bridge using forced oscillation method. *J. Wind Eng. Ind. Aerod.* 196, 104029.
 Wind-resistant design specification for railway bridges, Q/CR 9162-2023, 2023.
 Wang, Y., Guo, W., Xia, H., Wang, S., Xu, M., 2021. Coupling vibration response analysis of wind-train-bridge system considering the train-induced wind effect. *Wind Struct.* 33 (3), 217–231.
 Wu, F., Wang, Z., Zhao, L., Pan, T., Xiao, H., Ge, Y., 2023. Aerodynamic force distribution characteristics around a double-slotted box girder of a long-span bridge during vortex-induced vibration. *J. Bridge Eng.* 28 (1), 04022132.
 Xu, Y.L., Guo, W.H., 2004. Effects of bridge motion and crosswind on ride comfort of road vehicles. *J. Wind Eng. Ind. Aerod.* 92 (7–8), 641–662.
 Yan, Y., Yagi, T., Noguchi, K., Ito, Y., Shimada, R., 2022. Effects of handrail details on vortex-induced vibration for a Box-girder bridge. *J. Bridge Eng.* 27 (3), 04021114.
 Yu, H., Wang, B., Zhang, G., Li, Y., Chen, X., 2020. Ride comfort assessment of road vehicle running on long-span bridge subjected to vortex-induced vibration. *Wind Struct.* 31 (5), 393–402.
 Zhou, Y., Chen, S., 2016. Vehicle ride comfort analysis with whole-body vibration on long-span bridges subjected to crosswind. *J. Wind Eng. Ind. Aerod.* 155, 126–140.
 Zhu, Q., Xu, Y.L., Zhu, L.D., Li, H., 2018. Vortex-induced vibration analysis of long-span bridges with twin-box decks under non-uniformly distributed turbulent winds. *J. Wind Eng. Ind. Aerod.* 172, 31–41.
 Zhu, J., Xiong, Z., Xiang, H., Huang, X., Li, Y., 2021. Ride comfort evaluation of stochastic traffic flow crossing long-span suspension bridge experiencing vortex-induced vibration. *J. Wind Eng. Ind. Aerod.* 219, 104794.